

Vigneshwaran C. J.

AI / ML Engineer and Full-Stack Developer

Coimbatore, India | vigneshwarancj@gmail.com

<https://www.linkedin.com/in/vigneshwarancj1/> | <https://github.com/vigneshcj001>

PROFILE

Full-stack developer and AI / ML engineer with an M.Tech in Big Data Biology. Work spans enterprise applications, WhatsApp automation, real-time networking, and computational biology tools.

EXPERIENCE

Junior Software Developer | Ceiyone Tech Works Pvt. Ltd.

Dec 2025 - Present

Designing, developing, and maintaining enterprise-grade applications and AI-powered platforms - including AIORA, a multi-tenant WhatsApp business platform (bookings, storefronts, and n8n-driven AI conversations). Work spans backend APIs, WhatsApp automation, AWS cloud deployments, AI/LLM integration, and internal monitoring systems. Also mentoring 2 engineering interns on the AIORA platform.

AI Engineer Intern | Ceiyone Tech Works Pvt. Ltd.

Oct 2025 - Dec 2025

Built AI-powered automation solutions, OCR-based document systems, LinkedIn automation tools, and multi-agent LLM systems. Integrated AI services, external APIs, and workflow automation platforms.

Research Intern | Manipal Institute of Technology

Jun 2024 - Jul 2024

Developed ML-based QSPR models to predict the solubility of Posaconazole in Self-Emulsifying Drug Delivery Systems (SEDDS). Used RDKit for molecular descriptor generation (50+ descriptors), VIF analysis for feature selection (10 key descriptors: ELUMO, EHOMO, MaxQ-, MaxQ+, MinPartialCharge, MaxPartialCharge, MinAbsPartialCharge, FpDensityMorgan2, PEOE_VSA1, HallKierAlpha). Evaluated Linear Regression, Random Forest, PLS, KNN, and SVM models.

Tutor | Kamakshi Tuition Centre

Jan 2023 - Jun 2023

Taught mathematics to 9th grade students and Python programming to 12th grade students at Kamakshi Tuition Centre, Mugalivakkam Arasa Maram.

SELECTED PROJECTS

Syncly

Syncly is a full-stack professional networking platform Vigneshwaran built from scratch, designed to help individuals, researchers, and professionals across diverse sectors (Tech, Biology, Commerce, IT, and more) connect, collaborate, and share projects. Unlike traditional networking apps, Syncly emphasises real collaboration through swipe-based matchmaking (right swipe = Vibe to connect, left swipe = Ghost to skip), real-time chat via Socket.IO with typing indicators and active user tracking, a custom portfolio builder (users generate a professional responsive portfolio by filling out forms - no coding required), AWS SES integration for onboarding and notification emails with cron-based daily collaboration summaries, and scalable modular REST APIs. Deployed on AWS EC2.

Technologies: React.js, Redux, Tailwind CSS, Node.js, Express.js, MongoDB, Socket.IO, AWS EC2, AWS SES, Node-cron, Radix UI

GlycanBench: integrated resource for working with glycans

GlycanBench is a comprehensive glycoinformatics platform and Python package developed by Vigneshwaran CJ under the guidance of Ashok Palaniappan at the Systems Computational Biology Lab, SASTRA Deemed University. Glycans are monosaccharide polymers fundamental to life, playing pivotal roles in immunity and all major human diseases. Most host-pathogen interactions depend on glycan structures. GlycanBench covers the full glycan research workflow: creation (glycan molecules, biosynthetic networks, format conversion), visualisation (2D draw, 3D representation, KEGG pathway view), analysis (monosaccharide/glycan characterisation, descriptor generation, motif mutation), comparative analysis via molecular fingerprints (Morgan $r=2/3$, AtomPair, Torsion, RDKit), sequence alignment with custom scoring, hierarchical/k-means clustering with outlier detection, an optimised MPNN model for immunogenicity prediction with explainable outputs, and a glycobiology chat interface. All tools provide publication-quality visualisation. Freely available as a web-server and Python package for academic and non-profit use.

Technologies: Python, FastAPI, Flask, React.js, Tailwind CSS, PyTorch, MPNN, RDKit

EDUCATION

Master of Technology in Big Data Biology

SASTRA Deemed University | Aug 2023 - Jun 2025 | 79.67%

Bachelor of Technology

Prathyusha Engineering College | Aug 2019 - Apr 2023 | 84.4%

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, JavaScript, TypeScript

Data Science & ML: Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow, SHAP

Frontend Development: React.js, Next.js, Redux, HTML5, CSS3, Tailwind CSS, Bootstrap

Backend Development: Node.js, Express.js, Flask, FastAPI, Streamlit, Socket.IO, BullMQ

Databases: MongoDB, PostgreSQL, DynamoDB, Oracle, Redis

Cloud & Devops: AWS EC2, AWS SES, Google Cloud, Firebase

Ai & Llms: GPT-4, LangChain, LangGraph, Groq, Agentic AI

Bioinformatics & Research: RDKit, Mordred, Glycomics, Cytoscape, Galaxy

Tools: Git, GitHub, n8n, Postman, Jupyter Notebook, Linux, Vim, Arduino, Photoshop, macOS

CERTIFICATIONS

Namaste ReactJS | NamasteDev

Namaste Node.js | NamasteDev